

Internship — Project 2 Assignments

Machine Learning & Data Science | Algorithm Implementation

Student Name	Project Title	Algorithms
Amanullah Fazil S	House Price Prediction & Analysis	Linear Regression + Decision Tree
Udayanidhi S	Diabetes Prediction System	Logistic Regression + Random Forest
Muthu Sankar V	Flower Species Classification	KNN + SVM
Lakshana R	Titanic Survival Prediction	Decision Tree + Random Forest
M Mugesh	Chess Game Outcome & Move Prediction	Linear Regression + Random Forest
Nitesh Varshan K K	Heart Disease Risk Prediction	Logistic Regression + KNN
Palani Bharathi	Wine Quality Classification	Decision Tree + KNN
Nithish T	Breast Cancer Tumor Classification	SVM + Random Forest
S Ramkumar	Sales Prediction from Advertising Spend	Linear Regression + KNN

Each student is assigned 2 algorithms. Download the dataset from the provided Kaggle link, implement both algorithms, compare results, and submit code + report.

Amanullah Fazil S

Project Title	House Price Prediction & Analysis
Algorithms	Linear Regression Decision Tree
Dataset	California Housing Dataset

Objective

Build a predictive model to estimate house prices based on features such as median income, house age, average rooms, and location. Compare Linear Regression and Decision Tree models and derive insights from the data.

Tasks to Complete

- 1 Load and explore the California Housing Dataset. Check shape, info, describe, and null values.
- 2 Visualize feature correlations using a heatmap. Identify the top 3 features most correlated with price.
- 3 Preprocess: handle missing values and scale numerical features using StandardScaler.
- 4 Split data into 80% training and 20% testing sets.
- 5 Train a Linear Regression model. Evaluate using MAE, MSE, and R² score.
- 6 Train a Decision Tree Regressor. Tune max_depth (try 3, 5, 10) and report best results.
- 7 Plot Actual vs Predicted values for both models.
- 8 Compare both models in a summary table and write a short conclusion on which performed better and why.

Udayanidhi S

Project Title	Diabetes Prediction System
Algorithms	Logistic Regression Random Forest
Dataset	Pima Indians Diabetes Dataset

Objective

Build a classification model to predict whether a patient is diabetic based on medical attributes such as glucose level, BMI, age, and blood pressure. Implement and compare Logistic Regression and Random Forest classifiers.

Tasks to Complete

- 1 Load the dataset and perform EDA. Check for class imbalance in the target column.
- 2 Visualize distributions of key features (Glucose, BMI, Age) using histograms and box plots.
- 3 Handle zero values in columns like Glucose and BMI by replacing them with column median.
- 4 Split data into 80% train and 20% test. Apply feature scaling.
- 5 Train a Logistic Regression model. Report accuracy, precision, recall, and F1 score.
- 6 Train a Random Forest classifier (n_estimators=100). Report the same metrics.
- 7 Plot the confusion matrix for both models.
- 8 Display feature importance from the Random Forest model using a bar chart.
- 9 Write a comparison summary: which model is more suitable for medical prediction and why.

Muthu Sankar V

Project Title	Flower Species Classification
Algorithms	KNN SVM
Dataset	Iris Flower Dataset

Objective

Classify iris flowers into three species — Setosa, Versicolor, and Virginica — using petal and sepal measurements. Implement and compare K-Nearest Neighbors (KNN) and Support Vector Machine (SVM) classifiers.

Tasks to Complete

- 1 Load the Iris dataset. Explore class distribution and feature names.
- 2 Visualize data using a pair plot (seaborn.pairplot) to observe class separability.
- 3 Preprocess: apply StandardScaler to normalize features.
- 4 Split data into 75% train and 25% test.
- 5 Train a KNN classifier. Try $k = 3, 5, 7$. Report accuracy for each.
- 6 Train an SVM classifier using both linear and RBF kernels. Report accuracy for each.
- 7 Plot confusion matrices for the best KNN and best SVM models.
- 8 Write a summary: which model gave better accuracy and why SVM handles small datasets well.

Lakshana R

Project Title	Titanic Survival Prediction
Algorithms	Decision Tree Random Forest
Dataset	Titanic Dataset

Objective

Predict whether a Titanic passenger survived based on features like passenger class, age, gender, and fare. Build and compare Decision Tree and Random Forest classifiers.

Tasks to Complete

- 1 Load the Titanic dataset. Drop irrelevant columns (Name, Ticket, Cabin).
- 2 Handle missing values: fill Age with median, Embarked with mode.
- 3 Encode categorical features: convert Sex and Embarked using Label Encoding.
- 4 Visualize survival rate by gender, passenger class, and age group.
- 5 Split data into 80% train and 20% test.
- 6 Train a Decision Tree classifier. Visualize the tree (max_depth=3).
- 7 Train a Random Forest classifier (n_estimators=100). Report accuracy, precision, recall, F1.
- 8 Plot feature importance from the Random Forest model.
- 9 Compare both models and conclude which is more reliable for this dataset.

M Mugesh

Project Title Chess Game Outcome & Move Prediction

Algorithms Linear Regression | Random Forest

Dataset [Chess Games Dataset](#)

Objective

Analyze chess game data to predict the number of turns a game will last (regression) and classify the game winner (White/Black/Draw) based on features like player ratings, opening moves, and game type. Compare Linear Regression and Random Forest models on this real-world strategy game dataset.

Tasks to Complete

- 1 Load the Chess Games dataset. Explore columns: rated, turns, winner, white_rating, black_rating, opening_name, etc.
- 2 Perform EDA: plot distribution of game winners, average turns per result, and rating distributions for both players.
- 3 Encode categorical columns (winner, victory_status, opening_name) using Label Encoding.
- 4 Create a new feature: rating_diff = white_rating - black_rating. Analyze how it correlates with the winner.
- 5 Task A - Regression: Predict the number of turns. Split 80/20 and train a Linear Regression model. Evaluate using MAE, RMSE, and R2 score.
- 6 Task B - Classification: Predict the game winner. Train a Random Forest classifier (n_estimators=100). Report accuracy, precision, recall, and F1 score.
- 7 Plot Actual vs Predicted turns for the Linear Regression model.
- 8 Display the top 5 most important features from the Random Forest using a bar chart.
- 9 Write a conclusion: does player rating difference strongly decide the winner? Which model performed best?

Nitesh Varshan K K

Project Title	Heart Disease Risk Prediction
Algorithms	Logistic Regression KNN
Dataset	Heart Disease Dataset

Objective

Predict the presence or absence of heart disease in patients using clinical features like age, cholesterol, chest pain type, and maximum heart rate. Compare Logistic Regression and KNN classifiers.

Tasks to Complete

- 1 Load the Heart Disease dataset. Explore all features and check for missing values.
- 2 Visualize key features: age distribution, chest pain type counts, and correlation heatmap.
- 3 Encode categorical features and scale numerical features using StandardScaler.
- 4 Split data into 80% train and 20% test.
- 5 Train a Logistic Regression model. Report accuracy, precision, recall, and F1 score.
- 6 Train a KNN classifier. Try $k = 3, 5, 7, 9$. Select the best k using accuracy.
- 7 Plot confusion matrices for both models.
- 8 Plot the KNN accuracy curve (k vs accuracy) to visualize model selection.
- 9 Write a conclusion on which model is better suited for medical risk prediction.

Palani Bharathi

Project Title Wine Quality Classification

Algorithms Decision Tree | KNN

Dataset [Wine Quality Dataset](#)

Objective

Classify wines as good or bad quality based on physicochemical properties like acidity, sugar, pH, and alcohol content. Build and compare a Decision Tree classifier and a KNN classifier.

Tasks to Complete

- 1 Load the Wine Quality dataset. Understand all feature columns and quality score distribution.
- 2 Binarize the target: quality ≥ 7 as Good (1), else Bad (0).
- 3 Perform EDA: visualize feature distributions and class balance.
- 4 Scale features using StandardScaler. Split data 80/20 train/test.
- 5 Train a Decision Tree classifier. Visualize the tree (max_depth=4).
- 6 Train a KNN classifier (try $k = 3, 5, 7$). Report best accuracy.
- 7 Plot confusion matrix and classification report for both models.
- 8 Identify which chemical features most influence wine quality using the Decision Tree.
- 9 Summarize findings: which model gave better results and what quality factors matter most.

Nithish T

Project Title Breast Cancer Tumor Classification

Algorithms SVM | Random Forest

Dataset [Breast Cancer Wisconsin Dataset](#)

Objective

Classify breast tumors as malignant or benign using features derived from cell nucleus measurements. Build and compare SVM and Random Forest classifiers, and identify the most significant diagnostic features.

Tasks to Complete

- 1 Load the Breast Cancer dataset. Explore shape, feature names, and class distribution.
- 2 Perform EDA: compare mean feature values (radius, texture, smoothness) for both classes.
- 3 Apply StandardScaler to normalize features. Split data 80/20.
- 4 Train an SVM with RBF kernel. Report accuracy, precision, recall, and F1 score.
- 5 Train a Random Forest classifier (n_estimators=150). Report the same metrics.
- 6 Plot confusion matrices for both models side by side.
- 7 Display the top 10 most important features from Random Forest using a bar chart.
- 8 Use 5-fold cross-validation on both models and compare mean accuracy.
- 9 Write a conclusion on which features are most predictive of malignancy.

S Ramkumar

Project Title	Sales Prediction from Advertising Spend
Algorithms	Linear Regression KNN
Dataset	Advertising Dataset

Objective

Predict product sales based on advertising budgets across TV, Radio, and Newspaper channels. Build a Linear Regression model and a KNN Regressor, compare their performance, and determine which advertising channel has the strongest impact on sales.

Tasks to Complete

- 1 Load the Advertising dataset. Explore the 4 columns: TV, Radio, Newspaper, Sales.
- 2 Visualize relationships between each ad channel and sales using scatter plots.
- 3 Plot a correlation heatmap to identify which channel is most correlated with sales.
- 4 Split data into 80% train and 20% test.
- 5 Train a Linear Regression model. Print coefficients and interpret which channel matters most.
- 6 Train a KNN Regressor. Try $k = 3, 5, 7$. Choose the best k using RMSE.
- 7 Plot actual vs predicted sales for both models.
- 8 Evaluate both using MAE, RMSE, and R^2 score.
- 9 Write a business conclusion: where should the company invest their ad budget?

Submit completed project with: Python code (.ipynb or .py) | Output screenshots | Brief report (PDF/Word)